HAOTIAN (HIMMEL) XU

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EDUCATION

STONY BROOK UNIVERSITY, COLLEGE OF ENGINEERING AND APPLIED SCIENCES — Stony Brook, NY
Doctor of Philosophy in Data Science — May 2028

TEXAS A&M UNIVERSITY, COLLEGE OF ENGINEERING — College Station, TX
Master of Computer Science, MCS — May 2023

RUTGERS UNIVERSITY, SCHOOL OF ART & SCIENCE — New Brunswick, NJ
Bachelor of Science in Computer Science & Mathematics — Jan 2021
Honors in Computer Science

PUBLICATIONS

ICML 2026: "Resting Neurons, Active Insights: Robustifying Activation Sparsity in LLMs via Spontaneity", **Haotian Xu**, Tian Gao, Tsui-Wei Weng, Tengfei Ma

KDD 2026: "CAAD: Causality-Aware Multivariate Time Series Anomaly Detection via Multi-Scale Alignment and Structural Causal Consistency", Xin Wang, Yunshi Wen, Yanan He, **Haotian Xu**, youlan zhao, Michel Ferreira Cardia Haddad, Tengfei Ma

arXiv: "HyperGuide: Hyperbolic Guidance for Efficient Multi-Step Reasoning in Large Language Models", Yuyu Liu, **Haotian Xu**, Yanan He, Sarang Rajendra Patil, Mengjia Xu, Tengfei Ma

arXiv: "Concept-Guided Backdoor Attack on Vision Language Models", Haoyu Shen, Weimin Lyu, **Haotian Xu**, Tengfei Ma

TMLR 2026: "Graph Concept Bottleneck Models", **Haotian Xu**, Tsui-Wei Weng, Lam M. Nguyen, Tengfei Ma

LoG 2025: "When Structure Doesn't Help: LLMs Do Not Read Text-Attributed Graphs as Effectively as We Expected", **Haotian Xu**, Yuning You, Tengfei Ma

ICLR 2024: "Latent 3D Graph Diffusion", Yuning You, Ruida Zhou, Jiwoong Park, **Haotian Xu**, Chao Tian, Zhangyang Wang, Yang Shen

ICLR 2024 Workshop: "Multi-Modal Contrastive Learning for Proteins by Combining Domain-Informed Views", **Haotian Xu**, Yuning You, Yang Shen

NAACL 2024: "EventMoral: Moral Opinions towards Events in News Articles", Yuanyuan Lei, Ayesha Qamar, Jonathan Tong, **Haotian Xu**, Md Messal Monem Miah, Sai Ramana Reddy, Ruihong Huang

NAACL 2022: "Crossroads, Buildings and Neighborhoods: A Dataset for Fine-grained Location Recognition", Pei Chen, **Haotian Xu**, Cheng Zhang, Ruihong Huang

RESEARCH EXPERIENCE

SBU GRAPH LAB — Stony Brook, NY
Research Assistant — Aug 2023 – present

- **LLM Reasoning**: Implemented a self-distillation post-training method with integration of latent chain-of-thoughts and generative modeling; increased the efficiency of LLM inference with reasoning through hyperbolic space construction and adaptation, the final token consumption is less than 0.1% of tree-of-thoughts; under review at ICLR 2027.
- **Efficient LLM**: Designed a neuron-level LLM pruning and sparsified more than 60% neurons inside LLMs while maintaining comparable performance on 7 LM Evaluation Harness tasks; accepted by ICML 2026.
- **LVLM robustness**: Established a novel backdoor attack with consideration of activated visual concepts, and the attack success rate reached 100% on COCO and Flickr 8k datasets; under review at TMLR..
- **LLM Safety**: Extracted LLM circuits using input information and sparsified more than 95% of computation units inside LLMs while maintaining 77.72% accuracy on the Indirect Object Identification task and 65.59% on the Gendered Pronous task.
- **Graph-LLM**: Developed a novel Multimodal LLM to reason over graphs and point clouds; attained an average 90% link prediction performance on the ogbn benchmark; analyzed the effect of structural information in LLMs; accepted by LoG 2025.
- **Mechanistic Interpretability**: Enhanced the interpretability of VLMs like CLIPs, using advanced CBMs integrating with Graph Structure Learning; achieved on average 3% improvements on 6 different datasets; accepted by TMLR 03/2026.
- **LLM Post-Training**: Recovered the LoRA fine-tuning performance via light-weight steering vectors (require about 0.9% of LoRA params) and data-attribution (50% of LoRA trained data) on WinoGrande, SDF, and EM tasks (LoRA 59.0% v.s. ours 58.3%).

BIOLOGICAL INTELLIGENCE LAB AT TAMU — College Station, TX
Student Worker — Nov 2022 –Sep 2023

- **AI4Science**: Proposed a novel framework that learns to generate high-quality 3D molecular structures by diffusing over a latent graph representation, enabling efficient and accurate modeling of complex spatial and chemical relationships.
- **Multi-Modality Training**: Built a new protein multimodal encoder to obtain a 78.16% F-1 score on Enzyme Commission (EC) number prediction task and 52.96% F-1 score on Gene Ontology (GO) term prediction benchmark.
- **Graph Prompting**: Designed node prompt tuning achieved 82.01% F-1 score on EC task and saved 20% computational cost.
- **Computational Bioinformatics**: Applied AlphaSeek clustering method to downsample AlphaFold2 dataset from more than 1M protein instances to 55,173 datapoints, while the 55k sub-dataset can still act as a good pre-training dataset.

NATURAL LANGUAGE GROUP AT TAMU — College Station, TX
Student Worker — Oct 2021 –Sep 2023

- **Knowledge Aware Benchmark**: Presented a large-scale dataset of approximately 9,600 event-level moral judgments annotated across 45,000 events in 400 news articles, and demonstrates that incorporating these moral-opinion labels into models enhances performance on tasks like ideological bias detection and subjective event classification.
- **NER datasets**: Introduced HarveyNER, a richly annotated tweet dataset targeting precise geographic mentions and demonstrates that curriculum learning enhances NER model performance on these challenging, informal, often lengthy location expressions.

INDUSTRY EXPERIENCE

UBER TECHNOLOGIES, INC — Sunnyvale, CA
PhD Software Engineer Intern — June 2026 – August 2026

- **Receipt Transcription**: Built and productionized a low-latency agentic system for retail cart auditing; cut daily token usage ~80% (~$3.5K → <$0.8K/day, around $1M yearly) and P50 latency from P95 9 mins to P95 30 secs by optimizing token usage and fusing ~4.6 LLM calls into 2.26, while improving audit precision +2.8 pp. Designed a single-call OCR-free "nudge" pipeline (P95 < 6 s, ≥ 85% precision) for real-time in-store error correction. Both new pipelines have been in production.
- **Persona-Based Agentic GenAI Workflow**: Build an agentic generation workflow that integrates user-behavior signals, persona inference, visual understanding, and image generation to dynamically produce personalized store hero images for improved recommendation relevance and user engagement.
- **Casual Event Foundation Model Deployment**: Design and deployed a large-scale event foundation model over user chronological event histories, providing a reusable pre-trained backbone for downstream ranking, recommendation, and personalization models across the delivery platform.

MSK CANCER CENTER — New York City, NY
Graduate Research Assistant — Mar 2026 – May 2026

- **LVLM for Cancer Prediction**: Aims to deployed a concept-grounded, interpretable Large Vision-Language Model (LVLM) framework—leveraging multimodal pretraining, concept bottlenecks, and parameter-efficient fine-tuning—to enhance the accuracy and trustworthiness of AI-assisted lung cancer diagnosis from both medical images and clinical text.
- **Diffusion+OT for Medical Image Analysis**: Desires to nhanced the robustness and generalizability of longitudinal tumor monitoring by combining conditional diffusion models for clinically realistic image augmentation with optimal transport-based feature alignment to accurately capture distribution shifts across different treatment stages.

AMAZON, IPC – SIMULATION — Bellevue, WA
Data Scientist Intern — Jun 2022 – Sep 2022

- **LVA-Filtering**: Designed an advanced filtering algorithm aiming to boost the data efficiency while minimizing the stimulation accuracy loss; filtered out 50% of data instances and maintained the simulation performance with an accuracy loss of 0.01%.
- **Fulfill-By-Amazon**: Validated the practicality of existing filtering method in the third-party marketplace; optimized the method by enlarging the hypothesis space and enhanced the performance by 7% F1-score.
- **Output Alarm**: Constructed an output alarm system using SVM, clustering, and KNN to assist downstream teams in classifying miscalculated simulated outputs and to ameliorate data quality by backpropagating to the input-alarm system.

SKILLS

- **Statistics & Quantitative Skills**: Time Series Forecasting, Regression Analysis, Simulation, Monte Carlo simulations, Stochastic Calculus, Statistical Inference, Convex Optimization, Information Theory, Gradient-based Methods
- **Programming Languages**: Python, C++, SQL, Java, PyTorch, TensorFlow, Hugging Face, Scikit-learn
- **ML Frameworks**: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn
- **ML Techniques**: Representation Learning, Model Interpretability, Contrastive Learning, LLM Fine-tuning, Pruning & Sparsity

ADDITIONAL INFORMATION

- **Interests**: video games, diving (PADI AOW and AIDA 2 Star Freediver), tennis, riding.